

FFT Self-Gravity in the AthenaK Shearing Box: Implementation and Validation

`athenak-multigrid` tree, branch `multigrid`

August 3, 2026

Abstract

This document records the design, mathematics, implementation, and validation of an FFT-based self-gravity solver for AthenaK’s shearing box, added to the `athenak-multigrid` tree. The solver follows the Athena-C implementation in `selfg_fft_disk.c`: the shearing-box phase-shift method of Gammie (2001) and the open (vacuum) vertical boundary condition of Koyama & Ostriker (2009), built on the `kokkos-fft` library so the same code runs on CPU and GPU backends. Validation: the triply periodic solve satisfies the discrete Poisson equation to machine precision (3.8×10^{-14}); the shearing solve is exact at shear-periodic instants and converges at the remap interpolation order (3rd for `ppmx`) against analytic shearing-wave solutions; the open-boundary solve is exact in the interior and matches the analytic slab potential to 0.15%; a Jeans-wave evolution agrees with the existing multigrid solver to nine significant digits while running $3.5\times$ faster; and a self-gravitating shearing wave reproduces linear swing-amplification theory through the swing (peak amplification 3.62 vs. 3.605 predicted at 256^2), with the wave’s phase locking to the instantaneous shearing wavevector holding to 10^{-13} and the remaining L_2 difference shown to be hydrodynamic in origin — it is the same size in a control run with gravity switched off.

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1 Plan and scope

1.1 Goal and roadmap

The long-term goal is gas self-gravity in AthenaK shearing-box simulations *with mesh refinement*. The strategy, in order:

1. **(this work)** A uniform-grid FFT Poisson solver with shearing-periodic horizontal boundaries and either periodic or open vertical boundaries. FFTs handle the shear-periodic boundary condition exactly (a per-mode phase shift), which is why every published shearing-box self-gravity implementation (Gammie 2001; Kim, Ostriker & Stone 2002; Pencil; TIGRESS; NIRVANA) is FFT-based.
2. A swing-amplification physics test (linear theory of self-gravitating shearing waves, cf. Kim & Ostriker 2001) exercising the full hydrodynamics + gravity loop.
3. Multi-rank support via `MPI_Allgatherv` at a seam left in the gather/scatter steps (Sec. 3).
4. The refinement hybrid: FFT on the uniform root grid provides the solution and the boundary data consistent with shear-periodicity; refined interior patches are solved with the existing multigrid module using Dirichlet boundary values prolonged from the root solution. Nothing in the present solver is discarded in that design.

1.2 Version-1 scope

The solver currently *requires* (and enforces with fatal errors): a 3D mesh, a uniform grid (no SMR/AMR — AthenaK already forbids refinement with a shearing box), a single MPI rank, and (shear-)periodic x_1/x_2 boundaries. The vertical boundary is periodic (with mean-density subtraction) or open. Shearing and non-shearing cases share one code path: with shear rate $q = 0$ every shear-specific step reduces to the identity.

2 Numerical method

2.1 The discrete Poisson equation and its exact k-space inverse

The gravity source terms difference the potential with second-order central differences, so the solver targets the *finite-difference* Poisson equation

$$(L\Phi)_{ijk} \equiv \frac{\Phi_{i+1} - 2\Phi_i + \Phi_{i-1}}{\Delta x^2} + \frac{\Phi_{j+1} - 2\Phi_j + \Phi_{j-1}}{\Delta y^2} + \frac{\Phi_{k+1} - 2\Phi_k + \Phi_{k-1}}{\Delta z^2} = 4\pi G \rho_{ijk}, \quad (1)$$

not the continuum equation. On a periodic uniform grid the DFT modes $e^{ik_x x_i}$ are eigenvectors of each difference operator: applying the stencil to $e^{ik_x i \Delta x}$ and factoring out the mode gives the eigenvalue

$$D(\mathbf{k}) = \frac{2 \cos(k_x \Delta x) - 2}{\Delta x^2} + \frac{2 \cos(k_y \Delta y) - 2}{\Delta y^2} + \frac{2 \cos(k_z \Delta z) - 2}{\Delta z^2} \xrightarrow{k\Delta \rightarrow 0} -k^2. \quad (2)$$

The solve is therefore: FFT $4\pi G \rho$, divide each mode by $D(\mathbf{k})$, inverse FFT. Because D is the exact symbol of the same stencil later used to compute forces, the returned Φ satisfies Eq. (1) to machine precision — there is no iteration residual (contrast multigrid). The $\mathbf{k} = 0$ mode is set to zero, which is exactly equivalent to subtracting the mean density (the Jeans swindle, required for consistency of the periodic problem).

2.2 Shearing box: the phase-shift (rolled-frame) method

With background velocity $v_{y,0} = -q\Omega x$, the shearing-periodic boundary condition identifies

$$\rho(x, y, z, t) = \rho(x + L_x, y - q\Omega L_x t, z, t), \quad (3)$$

so the box is strictly periodic only at times $t_n = n T_{\text{sh}}$ with $T_{\text{sh}} = L_y / (q\Omega L_x)$. Let t_s be the time since the most recent t_n and $\theta \equiv q\Omega t_s$ (code variable `qomt`). In sheared coordinates $y' = y + \theta x$ the density *is* doubly periodic:

$$\rho'(x, y', z) = \rho(x, y' - \theta x, z) \quad \Longleftrightarrow \quad \text{“roll”}: \text{shift each } x\text{-column of content by } +\theta x \text{ in } y. \quad (4)$$

A rolled-frame mode $e^{i(k'_x x + k_y y')}$ is a shearing wave with *physical* radial wavenumber

$$k_x = k'_x + \theta k_y \quad (\text{index form: } k_x \Delta x = (i_p + \theta \frac{L_x}{L_y} j_p) \frac{2\pi}{N_x}), \quad (5)$$

with j_p the *signed* mode index. The pipeline is: roll $\rho \rightarrow$ FFT $\rightarrow$ divide by $D(k'_x + \theta k_y, k_y, k_z) \rightarrow$ inverse FFT $\rightarrow$ unroll Φ (shift by $-\theta x$).

A point worth recording: if the roll/unroll used *spectral* (Fourier) interpolation, the composite operation would invert Eq. (1) exactly at all times — the per-column shift is then a pure phase factor $e^{ik_y \theta x}$, under which the unsheared stencil pulls back exactly to the shifted-eigenvalue form of Eq. (5). All error at $\theta \neq 0$ therefore comes from the finite-order remap interpolation, and the solver is exact whenever $\theta = 0$ (verified below to 3×10^{-14}).

The remap is the conservative flux form shared with AthenaK’s orbital advection (`shearing_box/remap_fluxes`). the integer part of the shift is folded into a periodic-wrapped load, the fractional part $\epsilon \in (-1, 1)$ is applied as $q_j \rightarrow q_j - (F_{j+1} - F_j)$ with upwind flux kernels of first (`dc`), second (`plm`), or third (`ppmx`) order. As a *static* remap, donor-cell reduces to linear interpolation (2nd-order pointwise), `plm` is 2nd order with a smaller constant, and `ppmx` is 3rd order — exactly the orders measured in Sec. 5.

Ghost zones. The gravity source term needs one valid face-adjacent ghost layer of Φ . Interior meshblock faces come from the global solution; x_2 ghosts wrap periodically; the physical x_1 ghost planes use the shear-periodic identification: the inner ghost plane is the outer boundary column with content shifted by $+\theta L_x \pmod{L_y}$, and vice versa, using the same remap kernel.

2.3 Open vertical boundary: the two-solve method (Koyama & Ostriker 2009)

Fourier transforming horizontally, each mode obeys $(d^2/dz^2 - k_\perp^2)\hat{\Phi} = 4\pi G\hat{\rho}$, whose isolated (vacuum) solution is the screened kernel $\hat{\Phi}(z) = -\frac{2\pi G}{k_\perp} \int \hat{\rho}(z') e^{-k_\perp|z-z'|} dz'$. A vertically periodic solve instead delivers an infinite stack of image slabs separated by L_z :

$$S_p = \sum_{n=-\infty}^{\infty} e^{-k_\perp|z-z'-nL_z|} = \frac{e^{-k_\perp\zeta} + e^{-k_\perp(L_z-\zeta)}}{1-q}, \quad q \equiv e^{-k_\perp L_z}, \quad \zeta = z - z' \in [0, L_z). \quad (6)$$

A second solve on $4\pi G\rho e^{-i\pi z/L_z}$ shifts the vertical spectrum by half a wavenumber (half-integer harmonics = *antiperiodic* vertical extension), whose Green's function is the alternating stack

$$S_a = \sum_n (-1)^n e^{-k_\perp|z-z'-nL_z|} = \frac{e^{-k_\perp\zeta} - e^{-k_\perp(L_z-\zeta)}}{1+q}. \quad (7)$$

Both are combinations of the same two functions of ζ ; solving the 2×2 system for the weights that cancel the wraparound term and normalize the vacuum kernel gives

$$\boxed{\alpha_p = \frac{1}{2}(1 - e^{-k_\perp L_z}), \quad \alpha_a = \frac{1}{2}(1 + e^{-k_\perp L_z}), \quad \alpha_p S_p + \alpha_a S_a = e^{-k_\perp \zeta}.} \quad (8)$$

Every periodic image cancels identically; the result is the isolated-slab potential at the cost of one extra FFT pair. In k-space, solve A multiplies by $\alpha_p/D(k'_x, k_y, k_z^{\text{int}})$ and solve B by $\alpha_a/D(k'_x, k_y, k_z^{\text{half}})$ with $k_z^{\text{half}}\Delta z = (m + \frac{1}{2})2\pi/N_z$; the recombination is $\Phi = \Phi_A + \text{Re}[e^{+i\pi\tilde{z}/L_z}\Phi_B]$ with cell-centered $\tilde{z} = z - z_{\text{min}}$.

At $k_\perp = 0$ the weights become $(0, 1)$: the horizontally averaged density is carried entirely by the antiperiodic solve, which correctly reproduces the linear $|z|$ -type slab potential; no mean-density subtraction occurs in open mode, and the $D = 0$ singularity never arises. Matching Athena-C, the screening exponent uses the *continuum* $k_\perp = [(k_x\Delta x/\Delta x)^2 + (k_y\Delta y/\Delta y)^2]^{1/2}$ (with the shear-shifted k_x) rather than the discrete decay rate κ defined by $2[\cosh(\kappa\Delta z) - 1]/\Delta z^2 = k_{\perp, \text{eff}}^2$; the difference matters only where $e^{-k_\perp L_z}$ is already exponentially negligible. The x_3 ghost layer is filled by linear extrapolation in open mode (the layers touching it are excluded from “interior” error norms).

2.4 Cost and normalization

Periodic mode: one complex-to-complex FFT pair per solve. Open mode: two pairs. FFT plans are created once and reused; normalization is the NumPy convention (inverse transform carries $1/N$), passed explicitly. Kernel coefficients (Eqs. 2, 5) are computed on the fly inside the k-space multiply — they are time-dependent through θ , and recomputing costs less than storing. The solve runs once per Runge–Kutta stage, matching the multigrid convention, so the potential is consistent with the stage's density.

3 What was built

3.1 New and modified files

File	Content
src/gravity/fft_gravity.hpp	FFTGravitySolver
src/gravity/fft_gravity.cpp	solver implementation
src/gravity/gravity.{hpp,cpp}	solver switch, Gravity
src/driver/driver.cpp	one line: pgrav->Solve()
src/multigrid/multigrid.hpp	pre-existing fix: Smooth
src/pgen/tests/fft_poisson.cpp	validation problem
src/pgen/tests/swing.cpp	swing-amplification
CMakeLists.txt, src/CMakeLists.txt, config.hpp.in	build integration
inputs/tests/fft_poisson*.athinput, jeans_fft.athinput, swing_selfgrav.athinput	test inputs
validation/	this document, figure

[†] Multigrid::Smooth accessed the not-yet-defined MultigridDriver inside the class body; Apple clang 21 rejects this, so the template body was moved below the driver class definition. No behavior change; the multigrid Jeans test still passes.

3.2 Solver pipeline

FFTGravitySolver::Solve() executes, per RK stage:

1. **Gather**: active-zone density from all meshblocks into a global (N_z, N_y, N_x) device array (block offsets from logical locations), scaled by $4\pi G$.
2. **Roll**: conservative per-column y -shift by $+\theta x$ (Eq. 4); identity when $\theta = 0$.
3. **k-space solve**: periodic (one FFT pair) or open (two pairs, Sec. 2.3), with the shear-shifted kernel (Eq. 5).
4. **Unroll**: shift by $-\theta x$.
5. **Ghost planes**: shear-periodic x_1 ghost columns ($\pm\theta L_x$ shifts).
6. **Scatter**: into pgrav->phi (active zones + one ghost layer); the unchanged SelfGravity source term consumes it.

The multi-rank seam: steps 1 and 6 are the only places that assume the pack spans the mesh; an MPI_Allgatherv between local gather and roll (and a rank-local scatter) upgrades the solver without touching steps 2–5.

3.3 Runtime options

Block/parameter	Values	Default	Meaning
<gravity> solver	multigrid fft	multigrid	Poisson solver selection
<gravity> four_pi_G	real	-1 (unset)	$4\pi G$ (pgens may override)
<gravity> vert_bc	periodic open	periodic	vertical boundary
<gravity> fft_remap	dc plm ppmx	plm	y -remap order
<shearing_box> qshear, omega0	real	—	read if the block exists
<hydro_srcterms> self_gravity	bool	false	enable source term

Existing multigrid input files run unchanged (default `solver=multigrid`).

4 Building and compiling

Prerequisites (done once; already satisfied on this machine):

```
cd athenak-multigrid
git submodule update --init kokkos      # Kokkos 4.7.02, vendored
brew install fftw                       # host FFT backend for kokkos-fft
```

Configure and build (kokkos-fft v2.0.0 is fetched automatically by CMake; it consumes the in-tree Kokkos target and links FFTW on host backends, cuFFT/hipFFT on GPU backends):

```
mkdir build && cd build
cmake -D Athena_ENABLE_FFT=ON ..
make -j8
```

Without `-D Athena_ENABLE_FFT=ON` the code builds exactly as before; requesting `solver=fft` then aborts with an explanatory error.

5 Tests and results

All tests use the `fft_poisson` problem generator, which fills a deterministic density field, calls the gravity solve once, and prints error diagnostics (lines starting `# FFT-POISSON`). Initial conditions are selected by `<problem> profile = modes | shwave | slab` and are overridable on the command line. The companion Julia notebook (`validation/fft_selfgravity_tests.ipynb`) re-runs every test and checks the outputs independently of the C++ diagnostics.

5.1 Discrete consistency (periodic, no shear)

Multi-mode + Gaussian-blob density, 64^3 , 8 meshblocks:

```
./build/src/athena -i inputs/tests/fft_poisson.athinput
# FFT-POISSON ERRORS: max_rel_all= 3.757e-14  l2_rel_all= 1.216e-14
```

The 7-point Laplacian of the returned potential reproduces $4\pi G(\rho - \bar{\rho})$ at machine precision, confirming the k-space inversion, the multi-block gather/scatter, and the periodic ghost fill simultaneously.

5.2 Shear bookkeeping: exactness at shear-periodic instants

Same field in a $q = 1.5$, $\Omega = 1$ shearing box ($T_{\text{sh}} = 2/3$): solving at $t = T_{\text{sh}}$ (i.e. $\theta = 0$) returns `max_rel` = $3.0\text{e-}14$ — the entire shear code path (roll, shifted kernel, unroll, shear ghosts) collapses to the exact periodic solve, as it must. At a generic phase ($t_0 = 0.37$, $\theta = 0.555$) the residual is finite and dominated by remap interpolation, as designed; see the caveat below. Figure 1 sweeps the shear phase across a full period: the residual drops by twelve orders of magnitude precisely at $t_0 = 0$ and $t_0 = T_{\text{sh}}$ and is flat in between, confirming that the modular shear-phase arithmetic ($\theta = q\Omega[t \bmod T_{\text{sh}}]$) has exactly the right period and that no error accumulates with time.

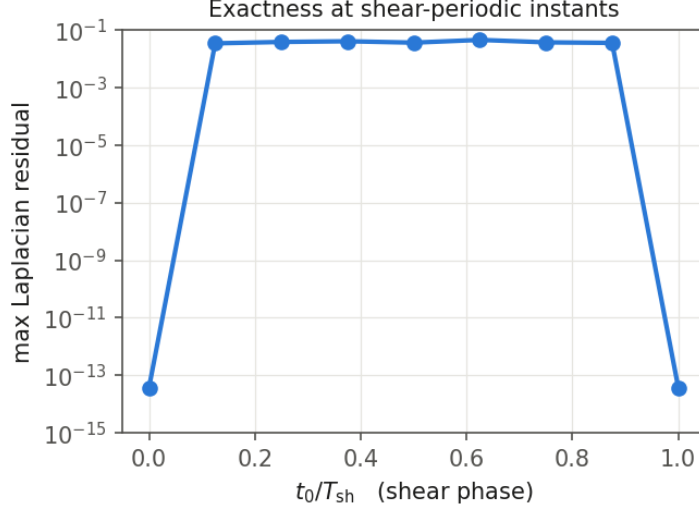


Figure 1: Laplacian residual versus shear phase over one shear period (64^3 , `modes` profile, `ppmx` remap). The solve is exact to machine precision exactly when the box is strictly periodic.

5.3 Shearing-wave solutions: order-of-accuracy

For a single rolled-frame mode (n_1, n_2, n_3) the exact discrete solution is known analytically: the same slanted wave with amplitude $4\pi G \delta\rho/D$, D evaluated with the shifted k_x (Eq. 5). Figure 2 shows the pointwise potential error at $\theta = 0.555$. Measured orders per resolution doubling: `ppmx` 3.08, 3.02; `p1m` 2.02, 2.01; `dc` 1.94, 2.05 — precisely the remap interpolation orders. Figure 3 shows the slanted wave and the solved potential (correlation coefficient -1.0 to 10^{-11} : potential wells align with density crests).

A caveat on metrics (important when re-validating). Under shear, the *Laplacian residual* of the returned potential is *not* a clean convergence metric: differencing divides the $O(\Delta y^p)$ remap interpolation error by Δ^2 , so the residual improves only at roughly first order (and the max-norm can stall at under-resolved sharp features, where the `ppmx` limiter locally reduces order). This is intrinsic to the method — with spectral interpolation the residual would be exactly zero — and is why the pointwise- Φ comparison of Fig. 2 is the correct order-of-accuracy test. The printed `max_rel_int/12_rel_int` norms exclude boundary layers whose ghosts are themselves interpolated (shear x_1 layers, extrapolated open-BC x_3 layers).

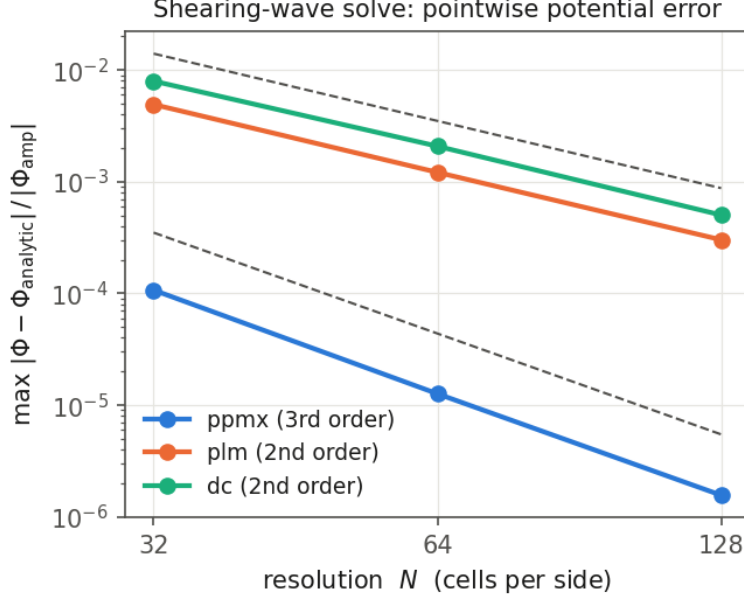


Figure 2: Maximum pointwise error of Φ against the analytic discrete shearing-wave solution, mode $(1, 1, 1)$, $q\Omega t_s = 0.555$, versus resolution, for the three remap orders. Dashed guides: N^{-2} , N^{-3} .

5.4 Open vertical boundary: slab potential

A horizontally uniform $\rho = \rho_0 \text{sech}^2(z/H)$ slab with $H = L_z/20$ at 64^3 , `vert_bc=open`: interior residual $4.2\text{e-}14$ (machine precision — for this profile only the $k_\perp = 0$ column is active, where the two-solve weights are exact), and the $\Phi(z)$ profile matches the analytic infinite-slab potential $4\pi G\rho_0 H^2 \ln \cosh(z/H)$ to 0.15% (Fig. 4); the residual peaks at the slab center (finite-difference discretization of the sharp maximum) and is limited by the truncated sech^2 wings at the box scale.

5.5 End-to-end evolution: Jeans wave, FFT vs. multigrid

The existing Jeans-wave test (`pgen_name=gravity`) evolved to $t = 1$ (321 cycles, RK2 $\Rightarrow$ 642 Poisson solves, 128 meshblocks of 8^3):

	mode amplitude at $t=1$	wall time	zone-cycles/s
FFT (<code>jeans_fft.athinput</code>)	$-7.395719378 \times 10^{-7}$	6.6 s	3.2×10^6
Multigrid (<code>jeans_wave.athinput</code>)	$-7.395719385 \times 10^{-7}$	22.9 s	0.9×10^6

Agreement to nine significant digits (the difference is the multigrid iteration residual; the FFT solve is exact), with a $3.5\times$ end-to-end speedup on this serial-CPU build. This exercises the full loop: per-stage solve, source-term coupling, and boundary/ghost handling during evolution.

5.6 Swing amplification: gravity coupled to the dynamics

All tests above check the Poisson solve in isolation. This one closes the loop: solver $\rightarrow$ source term $\rightarrow$ hydrodynamics, in a shearing frame.

A leading shearing wave ($k_{x0} < 0$, $k_y > 0$, $k_z = 0$) of small amplitude ($A = 10^{-4}$) is wound up by the background shear, $k_x(t) = k_{x0} + q\Omega k_y t$; as it swings through $k_x = 0$ self-gravity transiently

Shearing wave (rolled-frame mode $n = (1, 1, 1)$) at $q\Omega t_s = 0.555$: density and solved potential, $z = 0$ slice

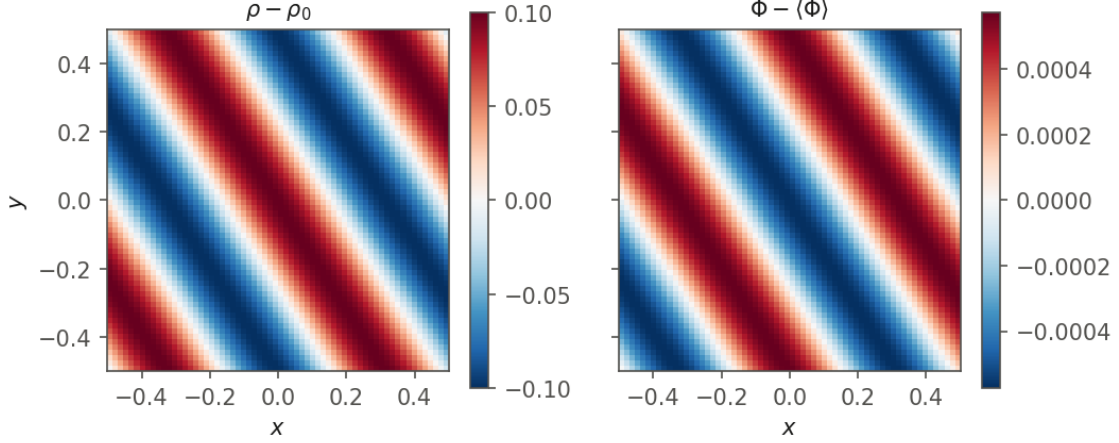


Figure 3: $z = 0$ slice of the shearing-wave test at 64^3 : density perturbation (left) and solved potential (right). The wavefronts are slanted by the accumulated shear; Φ is exactly anticorrelated with $\delta\rho$.

amplifies it (Goldreich & Lynden-Bell 1965; Julian & Toomre 1966; Toomre 1981). Linearizing the isothermal shearing-box equations for one shearing wave, with $\delta = \delta\rho/\rho_0$ and $w_{x,y} = i\hat{v}_{x,y}$ (so $v = w \sin \mathbf{k} \cdot \mathbf{x}$ while $\delta \propto \cos$):

$$\delta' = -(k_x w_x + k_y w_y), \quad w'_x = 2\Omega w_y + k_x S, \quad w'_y = -(2 - q)\Omega w_x + k_y S, \quad S = \delta \left(c_s^2 + \frac{4\pi G \rho_0}{D} \right), \quad (9)$$

where D is the solver's own discrete eigenvalue (Eq. 2, $\rightarrow -k^2$ in the continuum). Eq. (9) is derived from the shearing-box equations in Appendix A; setting $k_y = 0$ there recovers the Jeans dispersion relation $\delta'' = -(k^2 c_s^2 - 4\pi G \rho_0) \delta$, one of three checks collected in A.6.

Parameters are chosen in the epicyclic-resonant regime where swing is strong: $L_x = L_y = 2\pi$, $n_{wy} = 1$ so $k_y = 1 = \kappa$ (with $\kappa^2 = 2(2 - q)\Omega^2 = 1$ at $q = 1.5$), and $4\pi G \rho_0 = 1.9$, just below the marginal value $k_y^2 c_s^2 + \kappa^2 = 2$ — so the swinging wave is strongly amplified while every box mode remains Jeans-stable. Starting from $n_{wx} = -6$ the swing occurs at $t_{\text{swing}} = -n_{wx}/(q n_{wy}) = 4$, and the run covers $t \in [0, 8]$ (12 shear periods). The **swing** problem generator projects the solution onto the *instantaneous* wavevector each history dump, giving δ , w_x , w_y directly.

Results (Fig. 5): the simulation tracks the linear solution through the entire swing, reaching a peak amplification of 3.51 against the predicted 3.60, with relative L_2 differences of 9×10^{-3} (δ), 8×10^{-3} (w_x) and 1.3×10^{-2} (w_y) over the whole run at 128^2 . Two sharper diagnostics:

- The $k_x = 0$ crossing recorded in the history file occurs at $t = 4.005$ against the predicted 4.000 — the shear-phase bookkeeping is right after twelve shear periods.
- The *sine* projection of the density stays at $|\delta_{\text{sin}}|/|\delta|_{\text{max}} \sim 10^{-13}$ throughout. Linear theory says the wave stays exactly in phase with $\cos(\mathbf{k}(t) \cdot \mathbf{x})$; the simulation reproduces that phase locking to machine precision, which is a far more stringent statement than the amplitude agreement.

The residual $\sim 1\%$ difference is hydrodynamic truncation error, not a gravity error. Table 1 makes that quantitative by repeating the resolution study with self-gravity disabled, comparing each run against its own linear theory ($4\pi G = 1.9$ and $4\pi G = 0$ respectively). Two things follow:

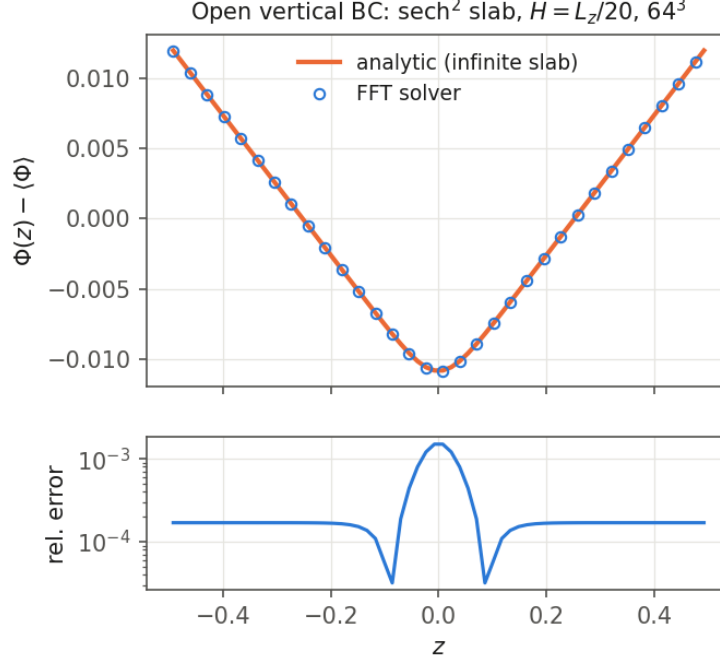


Figure 4: Open-BC slab test: solved $\Phi(z)$ (circles) versus the analytic infinite-slab potential (line), both zero-mean; lower panel: pointwise relative error.

- The peak amplification converges cleanly onto the predicted value, $2.64 \rightarrow 3.51 \rightarrow 3.62$ against 3.605 (a 0.4% difference at 256^2), while the control converges onto its own, much smaller, gravity-free value (1.80 vs. 1.49 predicted — the control is *not* amplified).
- At 256^2 the two L_2 errors are equal to within 3% (4.10×10^{-3} with gravity, 3.99×10^{-3} without). An error that is the same size whether or not the Poisson solver is running cannot originate in the Poisson solver: it is the PLM/HLLE/RK2 discretization of the hydrodynamics, which the linear ODE does not model. (The gravity-on sequence is non-monotonic in apparent order only because its 64^2 point is badly under-resolved — the peak amplitude is off by 27% there.)

Table 1: Swing solution versus linear theory (relative L_2 over $t \in [0, 8]$, each run compared against theory with its own $4\pi G$).

resolution	self-gravity ON		self-gravity OFF (control)	
	peak δ/δ_0	L_2 error	peak δ/δ_0	L_2 error
$64^2 \times 16$	2.638	9.08×10^{-2}	1.275	9.78×10^{-2}
$128^2 \times 16$	3.511	8.97×10^{-3}	1.734	1.81×10^{-2}
$256^2 \times 16$	3.619	4.10×10^{-3}	1.801	3.99×10^{-3}
linear theory	3.605	—	1.490	—

5.7 Reproducing

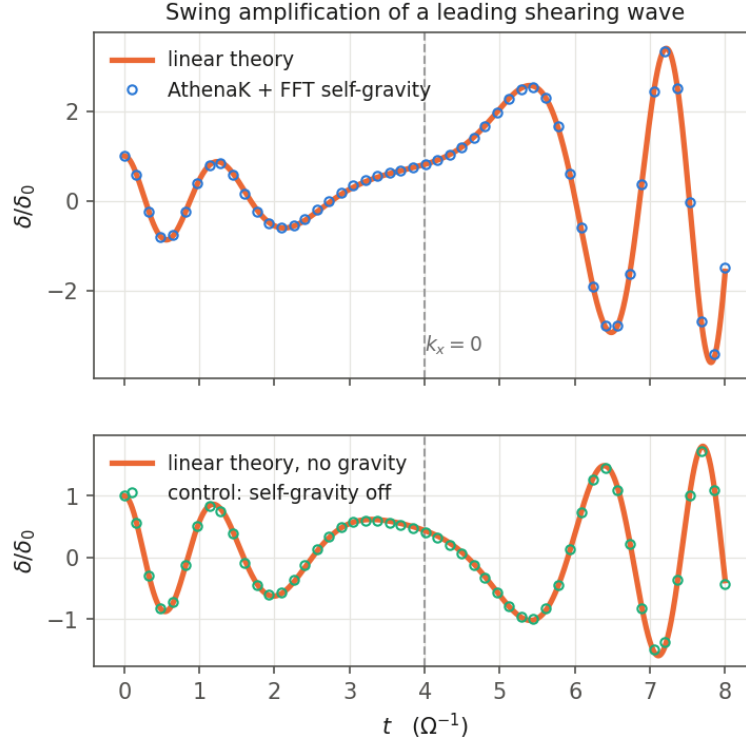


Figure 5: Swing amplification at $128^2 \times 16$. Top: density amplitude of the shearing wave versus time, simulation (circles) against the linear ODE (Eq. 9, line); dashed line marks $k_x = 0$. Bottom: the same run with self-gravity disabled, against the $4\pi G = 0$ theory — no amplification.

```
# periodic consistency
./build/src/athena -i inputs/tests/fft_poisson.athinput
# shear: generic phase / periodic instant
./build/src/athena -i inputs/tests/fft_poisson_shear.athinput
./build/src/athena -i inputs/tests/fft_poisson_shear.athinput \
    problem/time0=0.6666666666666667
# shearing-wave convergence (vary N, gravity/fft_remap, problem/n1..n3)
./build/src/athena -i inputs/tests/fft_poisson_shear.athinput \
    problem/profile=shwave gravity/fft_remap=ppmx \
    mesh/nx1=64 mesh/nx2=64 mesh/nx3=64 \
    meshblock/nx1=64 meshblock/nx2=64 meshblock/nx3=64
# open-BC slab
./build/src/athena -i inputs/tests/fft_poisson_open.athinput
# Jeans evolution
./build/src/athena -i inputs/tests/jeans_fft.athinput
# swing amplification (writes SwingSG.user.hst); control run without gravity:
./build/src/athena -i inputs/tests/swing_selfgrav.athinput
./build/src/athena -i inputs/tests/swing_selfgrav.athinput \
    job/basename=SwingNoG128 hydro_srcterms/self_gravity=false
```

Each run writes `bin/*.bin` snapshots (`hydro_w`, `grav_phi`) that the Julia notebook reads. Figures in this document are regenerated by `validation/figures/make_figures.py`.

6 Next steps

1. Multi-rank support (`MPI_Allgatherv` at the seam documented in Sec. 3), lifting the single-rank restriction.
2. Python regression harness under `tst/test_suite/fft_gravity/` mirroring the multigrid suite, so these checks run in CI.
3. The FFT-root + multigrid-refinement hybrid for the AMR goal: the FFT solve on the uniform root grid supplies both the global solution and, by prolongation, Dirichlet boundary data for multigrid on refined interior patches — which never touch the shear-periodic boundary.
4. Physics extensions once the above land: star-particle deposition into the density field (skipped in the Athena-C shearing branch, see Sec. 3), and stratified shearing boxes combining `vert_bc=open` with vertical gravity.

A Derivation of the linearized shearing-wave system, Eq. (9)

This appendix derives Eq. (9) — the reference solution against which the swing test of Sec. 5.6 is compared — from the shearing-box equations. Three checkpoints for verifying the algebra are collected in A.6.

A.1 Shearing-box equations and the background state

The shearing box is a local Cartesian frame co-rotating at radius R_0 with angular velocity $\boldsymbol{\Omega} = \Omega \hat{z}$; coordinates (x, y, z) are (radial, azimuthal, vertical) with $x = R - R_0$. For an isothermal, unstratified gas ($P = c_s^2 \rho$, no vertical external gravity):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{c_s^2}{\rho} \nabla \rho - \nabla \Phi \underbrace{- 2\boldsymbol{\Omega} \times \mathbf{v}}_{\text{Coriolis}} + \underbrace{2q\Omega^2 x \hat{x}}_{\text{tidal}}, \quad (10)$$

with $\nabla^2 \Phi = 4\pi G \rho$ and $q \equiv -d \ln \Omega / d \ln R$ ($q = 3/2$ Keplerian). The tidal term is the first-order expansion in x of the combined centrifugal and central-gravity forces; the Coriolis term written out is $-2\Omega \hat{z} \times \mathbf{v} = 2\Omega v_y \hat{x} - 2\Omega v_x \hat{y}$.

The background is uniform density in shear flow,

$$\rho = \rho_0, \quad \mathbf{v}_0 = -q\Omega x \hat{y}, \quad \Phi_0 = 0. \quad (11)$$

That this solves Eq. (10) is worth checking explicitly, because it fixes the sign convention for everything that follows: the x -momentum equation reduces to $0 = 2\Omega v_{0y} + 2q\Omega^2 x = -2q\Omega^2 x + 2q\Omega^2 x = 0$, i.e. the Coriolis force on the shear flow exactly cancels the tidal force. Setting $\Phi_0 = 0$ despite $\rho_0 \neq 0$ is the Jeans swindle; the solver enforces precisely this by zeroing the $\mathbf{k} = 0$ Fourier mode (Sec. 2.2).

A.2 Linearization

Write $\rho = \rho_0(1 + \delta)$, $\mathbf{v} = \mathbf{v}_0 + \mathbf{v}'$, $\Phi = \Phi'$, and define the background-advection operator

$$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + (\mathbf{v}_0 \cdot \nabla) = \frac{\partial}{\partial t} - q\Omega x \frac{\partial}{\partial y}. \quad (12)$$

Keeping terms of first order in $(\delta, \mathbf{v}', \Phi')$, and using $\nabla \cdot \mathbf{v}_0 = \partial_y(-q\Omega x) = 0$:

$$\frac{D\delta}{Dt} = -\nabla \cdot \mathbf{v}', \quad (13)$$

$$\frac{Dv'_x}{Dt} = -c_s^2 \frac{\partial \delta}{\partial x} - \frac{\partial \Phi'}{\partial x} + 2\Omega v'_y, \quad (14)$$

$$\frac{Dv'_y}{Dt} = -c_s^2 \frac{\partial \delta}{\partial y} - \frac{\partial \Phi'}{\partial y} - (2 - q)\Omega v'_x. \quad (15)$$

The only non-obvious term is the last one. The advection of the *background* azimuthal velocity by the radial perturbation contributes $(\mathbf{v}' \cdot \nabla)v_{0y} = v'_x \partial_x(-q\Omega x) = -q\Omega v'_x$ on the left-hand side; moving it right turns the bare Coriolis coefficient -2Ω into $-(2 - q)\Omega$. This $(2 - q)\Omega$ is the fingerprint of working with velocity *fluctuations* rather than total velocity, and it is exactly the coefficient appearing in AthenaK's `shearing_box_srcterms.cpp` — which is how the convention assumed here was confirmed against the code.

A.3 The shearing-wave ansatz

Because D/Dt carries an explicit x -dependent coefficient, plane waves of fixed wavevector are not solutions. Try instead a wave whose radial wavenumber depends on time,

$$f(\mathbf{x}, t) = \hat{f}(t) e^{i[k_x(t)x + k_y y + k_z z]} \implies \frac{Df}{Dt} = \left[\frac{d\hat{f}}{dt} + i \left(\frac{dk_x}{dt} - q\Omega k_y \right) x \hat{f} \right] e^{i\mathbf{k} \cdot \mathbf{x}}. \quad (16)$$

The residual x -dependence cancels if and only if

$$\frac{dk_x}{dt} = q\Omega k_y \implies k_x(t) = k_{x0} + q\Omega k_y t, \quad (17)$$

after which $D/Dt \rightarrow d/dt$ acting on the amplitudes. The shearing wave is therefore *forced* by the structure of the operator rather than assumed. Eq. (17) is the same relation that appears in the solver's k -space kernel (Eq. 5), in the `swing` history projection, and in AthenaK's pre-existing `shwave.cpp`.

A.4 Amplitude equations and the Poisson closure

Substituting $\nabla \rightarrow i\mathbf{k}(t)$ and specializing to $k_z = 0$ (so that $\hat{v}_z$ decouples and remains zero):

$$\frac{d\hat{\delta}}{dt} = -i(k_x \hat{v}_x + k_y \hat{v}_y), \quad \frac{d\hat{v}_x}{dt} = 2\Omega \hat{v}_y - ik_x S, \quad \frac{d\hat{v}_y}{dt} = -(2 - q)\Omega \hat{v}_x - ik_y S, \quad (18)$$

with the Poisson closure

$$S \equiv c_s^2 \hat{\delta} + \hat{\Phi}, \quad \hat{\Phi} = \frac{4\pi G \rho_0 \hat{\delta}}{D} \implies S = \hat{\delta} \left(c_s^2 + \frac{4\pi G \rho_0}{D} \right). \quad (19)$$

Here D is deliberately the *discrete* eigenvalue of Eq. (2) rather than its continuum limit $-k^2$, so that the reference solution inverts the same operator the solver does; the difference is $O((k\Delta x)^2)$ and is not negligible at the resolutions used. Note $D < 0$ for every non-zero mode, so self-gravity always *reduces* the restoring force, as it must.

One geometric caveat: with $k_z = 0$ in a three-dimensional periodic box the closure is $\hat{\Phi} = -4\pi G \rho_0 \hat{\delta} / k_\perp^2$, which is *not* the razor-thin sheet result $\hat{\Phi} = -2\pi G \hat{\Sigma} / |k_\perp|$ used in much of the swing-amplification literature. Both are correct in their own geometry; the test here is a 3D box with a vertically uniform perturbation, so the former applies.

A.5 Real form

Finally, substituting $w_x \equiv i\hat{v}_x$, $w_y \equiv i\hat{v}_y$ removes every explicit i and yields Eq. (9):

$$\delta' = -(k_x w_x + k_y w_y), \quad w'_x = 2\Omega w_y + k_x S, \quad w'_y = -(2 - q)\Omega w_x + k_y S. \quad (20)$$

Because the coefficients are real, real initial data stays real for all time, which has a directly testable consequence: the density perturbation remains exactly in phase with $\cos(\mathbf{k}(t) \cdot \mathbf{x})$ and the velocities with $\sin(\mathbf{k}(t) \cdot \mathbf{x})$, with no phase drift. This is why the **swing** history function projects density onto the cosine and velocity onto the sine of the instantaneous phase, and why the measured sine component of the density (Sec. 5.6, $\sim 10^{-13}$) is such a stringent check.

A.6 Checks on the algebra

Three limits verify the system independently:

1. **Jeans limit** ($k_y = 0$, $\Omega = 0$): eliminating w_x gives $\delta'' = -(k^2 c_s^2 - 4\pi G \rho_0)\delta$, the Jeans dispersion relation $\omega^2 = k^2 c_s^2 - 4\pi G \rho_0$. The companion notebook integrates the ODE in this limit and reproduces the analytic $\cos(\omega t)$ / $\cosh(|\omega|t)$ solutions to six decimal places in both the stable and unstable regimes.
2. **Epicyclic limit** ($S = 0$): $w''_x = -\kappa^2 w_x$ with $\kappa^2 = 2(2 - q)\Omega^2$, the epicyclic frequency ($\kappa = \Omega$ for $q = 3/2$). Obtaining $\kappa^2 = 4\Omega^2$ instead means the $(\mathbf{v}' \cdot \nabla)v_{0y}$ term in Eq. (15) was dropped.
3. **Marginal stability at swing** ($k_x = 0$): the tightly-wound dispersion relation $\omega^2 = k_y^2 c_s^2 - 4\pi G \rho_0 + \kappa^2$ sets the parameter choice of Sec. 5.6 — $4\pi G \rho_0 = 1.9$ against the marginal value $k_y^2 c_s^2 + \kappa^2 = 2$.